Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/31/2012
PI Name:	Bailey, Susan M. Ph.D.		
Project Title:	Telomeric proteins in the radiation/DNA damage response		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Colorado State University		
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PI Web Page:			
City:	Fort Collins	**State:**	CO
Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	01/01/2008	**End Date:**	12/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date is now 12/31/2013 per NSSC information (Ed., 9/27/2012) NOTE: End date is now 12/31/2012 per NSSC information (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liber, Howard (Colorado State University)		
Grant/Contract No.:	NNX08AB65G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research addresses Radiation Health Bioastronautics Roadmap (BR) Carcinogenesis Questions 28c, d and j, and is aimed at advancing current knowledge and mechanistic understanding of the DNA damage processing that underlies radiation effects in general, and space radiation in particular. The goal of the studies proposed here is to provide information that will aid identification of susceptible individuals, ultimately improving risk prediction for radiation carcinogenesis in astronauts exposed to high energy charged particles in space. Summary: Appropriate cellular response to natural chromosomal termini (telomeres) and to exogenously created double-strand breaks (DSBs) is critical to maintaining genomic stability. Incorrect handling of DSB ends leads to mutation, chromosomal rearrangements, instability and carcinogenesis. Altered levels of telomeric proteins have been reported in some tumors. The interplay between telomeres and DSBs may be better understood not only by viewing the striking differences between them, but perhaps even more importantly by viewing the obvious similarity-they are both DNA ends, the very substrate both telomere and damage response/repair proteins specifically recognize and bind. We have been investigating the role DNA repair proteins play at telomeric ends. Recent and intriguing studies suggesting that the classic telomere-binding protein TRF2 is also found (rapidly and transiently) at sites of photo-induced DNA DSBs, has turned the question around-what are telomere proteins doing at DSB ends? Here, we propose to address this fascinating question by investigating the role(s) telomeric proteins play in ionizing radiation (IR)-induced DNA damage response/processing/repair. Our approach assesses mutation frequency and spectra at the molecular level, as well as extends the analyses to the cytogenetic level. Our preliminary studies lend support to the view that telomeric proteins aren't just for telomeres anymore. Increasing our understanding of previously unrecognized relationships between the radiation/DNA damage response and telomere function will facilitate predicting the carcinogenic risk of exposure to radiation in general, and to astronauts exposed to HZE particles in particular. Ultimately, we foresee utilizing telomere instability/dysfunction as a useful biomarker of underlying repair deficiency in order to identify individual susceptibility to radiation carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our investigation of interdependencies between DNA repair and telomeres will ultimately lead to identification and characterization of previously unappreciated telomere dysfunction, which in turn will provide a useful indication of an underlying repair defect. Therefore, telomere instability may ultimately serve as a sensitive biomarker, identifying individual susceptibility to radiation carcinogenesis in general and individual susceptibility to HZE exposure in particular.
Task Progress:	Appropriate cellular response to natural chromosomal termini (telomeres) and to exogenously created double-strand breaks (DSBs) is critical to maintaining genomic stability. Incorrect handling of DSB ends leads to mutation, chromosomal rearrangements, instability and carcinogenesis. The interplay between telomeres and DSBs may be better understood not only by viewing the striking differences between them, but perhaps even more importantly by viewing the obvious similarity – they are both DNA ends, the very substrate both telomere and damage response/repair proteins specifically recognize and bind. We have been investigating the role DNA repair proteins play at telomeric ends. Studies suggesting that the “classic” telomere-binding protein TRF2 is also found (rapidly and transiently) at sites of photo-induced DNA DSBs turned the question around – what are telomere proteins doing at DSB ends? We have shown that ionizing radiation (IR)-induced DSBs are insufficient to initiate TRF2 recruitment to damage sites. We continue to address this fascinating question by investigating the role(s) telomeric proteins play in IR-induced DNA damage response/processing/repair. We have demonstrated that telomere protein deficiencies (TRF1, TRF2, POT1) significantly elevate spontaneous and IR-induced MFs, particularly associated with HZE exposure. • Identified novel aspect of telomere/DNA damage response/cancer risk interdependency. We have demonstrated that telomere associated tankyrase 1 normally serves to suppresses telomeric recombination (observed as T-SCE) in telomerase deficient backgrounds. Modeling suggests that elevated T-SCE frequencies accelerate senescence. • Identified novel aspect of telomere regulation with implications for IR-exposure & aging. We have demonstrated that telomere associated tankyrase 1 PARP activity serves to protect the NHEJ protein DNA-PKcs from proteasome-mediated degradation, likely explaining the radiosensitivity phenotype observed with depletion of tankyrase 1. Further, tankyrase PARP catalytic activity facilitates recruitment to and function of DNA-PKcs at DSB sites. • Identified novel mechanistic aspect of DNA-PKcs regulation with implications for DNA repair/cancer risk following exposure to ionizing radiations. Our results continue to provide support for the functional significance of telomere and IR-induced DNA damage response interdependencies - and the importance of individual genetic susceptibility considerations in ionizing radiation-induced cancer risk models. Therefore, these studies have enlightened models of cancer processes/risk in order to reduce uncertainty in radiation quality effects from GRC exposure. They have also enlightened models of cancer processes/risk in order to reduce uncertainty in individual radiation sensitivity (genetic factors) from GRC exposure. To further address these gaps, ongoing work is addressing other candidate protein involvement (e.g., telomerase and telomere length assessment) in critical cellular compartments (e.g., stem cells) for potential roles in telomere and/or DNA repair regulation, as well as to further investigate the effects of radiation quality at low doses to better understand cancer (and aging) processes induced by heavy ions.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
Articles in Peer-reviewed Journals	Chernikova SB, Razorenova OV, Higgins JP, Sishc BJ, Nicolau M, Dorth JA, Chernikova DA, Kwok S, Brooks JD, Bailey SM, Game JC, Brown JM. "Deficiency in mammalian histone H2B ubiquitin ligase Bre1 (Rnf20/Rnf40) leads to replication stress and chromosomal instability." Cancer Research. 2012 Apr 15;72(8):2111-9. Epub 2012 Feb 21. PubMed PMID: 22354749 , Apr-2012

Articles in Peer-reviewed Journals	Steffen LS, Bacher JW, Peng Y, Le PN, Ding LH, Genik PC, Ray FA, Bedford JS, Fallgren CM, Bailey SM, Ullrich RL, Weil MM, Story MD. "Molecular characterisation of murine acute myeloid leukaemia induced by 56Fe ion and 137Cs gamma ray irradiation." Mutagenesis.2012 Sep 17. [Epub ahead of print] PubMed PMID: 22987027 , Sep-2012
Articles in Peer-reviewed Journals	Loucas BD, Durante M, Bailey SM, Cornforth MN. "Chromosome damage in human cells by gamma-rays, a-particles and heavy ions: track interactions in basic dose response relationships." Radiation Research. In press, as of November 2012. , Nov-2012